

Restaurant Tips Prediction

Predicting Restaurant Tips Using Predictive Analytics in Excel

Objective

The objective of this project was to analyze a restaurant tips dataset and build a predictive model using Microsoft Excel to estimate tip amounts based on customer and dining characteristics.

Dataset Description

The dataset contained **244 customer records** with the following features:

Column	Description
sex	Gender of the customer
smoker	Whether the customer smokes
day	Day of the restaurant visit
time	Lunch or Dinner
size	Number of people dining
total_bill	Total bill amount in USD
tip	Tip amount in USD ← Target Variable

Tools & Techniques Used

Tool / Technique	Purpose
Microsoft Excel	Main analysis platform
Data Analysis ToolPak	Running regression model
AVERAGE, MAX, MIN, COUNT	Basic descriptive statistics
AVERAGEIF	Grouped analysis by day and smoker
Simple Linear Regression	Predictive modeling
Scatter Chart	Visualizing correlation
Bar Chart	Comparing average tips by group

Exploratory Data Analysis

Basic Statistics:

Metric	Value
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Average Tip	\$3.00
Highest Tip	\$10.00
Lowest Tip	\$1.01
Total Records	244

Average Tip by Day:

Day	Average Tip	Insight
Sunday	\$3.26	■ Highest
Saturday	\$2.99	Second highest
Thursday	\$2.77	Third
Friday	\$2.73	Lowest

Average Tip by Smoker Status:

Smoker	Average Tip
Yes	\$3.01
No	\$2.99

Key Finding: Smoking status had minimal effect on tipping behaviour. Day of visit and bill size were stronger predictors of tip amount.

Correlation Analysis

A scatter plot of Total Bill vs Tip revealed a **positive correlation** — as the bill amount increases, the tip amount also increases. The dots on the chart move upward from left to right, confirming that total bill is a strong predictor of tip amount.

Regression Model

Model Used: Simple Linear Regression

Total Bill was used as the predictor (X) variable and Tip as the target (Y) variable. The Data Analysis ToolPak in Excel was used to run the regression.

Regression Output:

Metric	Value	Meaning
R ² Score	0.4566	Bill explains 45.6% of tip variation
Intercept (β_0)	0.9203	Base tip before bill is considered
Coefficient (β_1)	0.1050	Every \$1 more bill = \$0.105 more tip

Regression Equation:

$$\text{Predicted Tip} = 0.92 + (0.105 \times \text{Total Bill})$$

Sample Predictions

Total Bill	Predicted Tip	Calculation
\$10.00	\$1.97	$0.92 + (0.105 \times 10)$
\$20.00	\$3.02	$0.92 + (0.105 \times 20)$
\$35.00	\$4.60	$0.92 + (0.105 \times 35)$
\$40.00	\$5.12	$0.92 + (0.105 \times 40)$
\$50.00	\$6.17	$0.92 + (0.105 \times 50)$

Key Insights

1. **Total bill is the strongest predictor** of tip amount — clear positive relationship
2. **Sunday generates the highest tips** — weekend diners are more relaxed and generous
3. **Smoking status barely affects tipping** — only \$0.02 difference
4. The regression model provides a **reliable baseline** for predicting tip amounts
5. **Higher bills = higher tips** — consistent throughout all 244 records

Conclusion

Using Simple Linear Regression in Excel, a predictive model was successfully built to estimate restaurant tips. The model equation **Predicted Tip = 0.92 + (0.105 x Total Bill)** provides a practical tool for predicting tip amounts based on bill size. The R^2 value of 0.4566 indicates a moderate fit, which is expected for human behaviour data where many external factors influence tipping decisions such as mood, service quality, and personal habits.